A Finite-Field Realization of the H_4 Coxeter Group, the 600-Cell, and the 24-Cell over $\mathbb{F}_{13}[\theta]$

Exact Golden-Ratio Arithmetic, No Real or Complex Numbers

Update for the State Space within Zeta: The First p-Adelic Integer Artificial Intelligence

Dávid Navrátil
david.navratil2016@gmail.com

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Abstract

This note builds and verifies the H_4/600-cell/24-cell structure entirely inside the finite field $\mathbb{F}_{169} = \mathbb{F}_{13}[\theta]/(\theta^2 - 5)$, using only exact modular integer arithmetic. No real square roots, no floating-point numbers, and no complex numbers appear anywhere in the mathematics below. Using the golden root φ of $x^2 - x - 1 = 0$ in this field, we build a finite-field analogue of the 120 standard 600-cell coordinates, verify its nine-class inner-product spectrum against the real H_4 cosines, construct genuine reflections, and verify by direct enumeration that the corresponding reflection group has order exactly 14400, matching the real H_4 Coxeter group. The conjugate root ψ generates a second, Galois-conjugate 120-point configuration sharing the same rational 24-cell core, with an explicit mirror map between the two. The integer companion matrix T_3 is attached to this same field through its Lagrange-Sylvester spectral projectors, computed by Bezout/CRT idempotents and lifted to $\mathbb{Z}_{13}$ by Hensel's lemma. We then classify every element of $\mathbb{F}_{169}^{\times}$ by multiplicative order (sixteen classes in total) and show that a single primitive root γ generates all of them: φ, ψ, the three eigenvalues of T_3, the extension generator θ, and a further order-24 element are all identified as explicit powers of γ. Finally, both 120-point configurations are shown to be genuine quaternion groups of order 120 (the binary icosahedral group $2I$), with the shared 24-point core a genuine subgroup of order 24 (the binary tetrahedral group $2T$), connecting this note to the classical H_4 folding of the E_8 root system.

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1 Introduction and motivation

The central object of the wider research programme is the integer Tribonacci companion matrix $T_3 \in SL(3,\mathbb{Z})$, whose reduction modulo 13 generates a cyclic subgroup of order 168 inside $SL(3,\mathbb{F}_{13})$. This note attaches a genuine, exactly verified H_4 Coxeter structure, the 600-cell and the 24-cell, to the same finite field that carries the spectral theory of T_3, and shows that the two objects share not only a field but a single discrete logarithm.

Everything here is built and checked inside $\mathbb{F}_{169} = \mathbb{F}_{13}[\theta]/(\theta^2 - 5)$, with no reference to real or complex numbers at any stage. This field exists as a genuine degree-2 extension of $\mathbb{F}_{13}$ because 5 is not a quadratic residue modulo 13:

$$\{x^2 \bmod 13 : x = 0, \ldots, 12\} = \{0, 1, 3, 4, 9, 10, 12\}, \qquad 5 \notin \text{this set}$$

(Figure 1). Every computation below is exact modular arithmetic on pairs (a, b) representing $a+b\theta$, $a, b \in \{0, \ldots, 12\}$; nothing is approximated.

2 The golden ratio in $\mathbb{F}_{169}$

A natural first candidate for a golden-ratio element is $\gamma = 7 + 8\theta$, since $7 \equiv 1/2 \pmod{13}$ in the real identity $\varphi = (1 + \sqrt{5})/2$. Checking this directly,

$$\gamma^2 - \gamma - 1 = (7 + 8\theta)^2 - (7 + 8\theta) - 1 = 10 \neq 0 \text{ in } \mathbb{F}_{169}.$$

Direct computation of the multiplicative order of $7 + 8\theta$ in $\mathbb{F}_{169}^{\times}$ gives order 168, so $\gamma = 7 + 8\theta$ generates the entire 168-element group rather than a 28-element golden-ratio cycle. It is not a root of $x^2 - x - 1$ at all. We return to γ in Section 7. Solving $x^2 - x - 1 = 0$ by exhaustive search over all 169 elements gives exactly two roots, $7 + 6\theta$ and $7 + 7\theta$, the two Frobenius-conjugate golden-ratio elements, matching the real identity $\varphi = (1 + \sqrt{5})/2$ with $1/2 = 7$ in $\mathbb{F}_{13}$. We fix $\varphi = 7 + 6\theta$ from here on. Table 1 gives its full cycle of 28 powers.

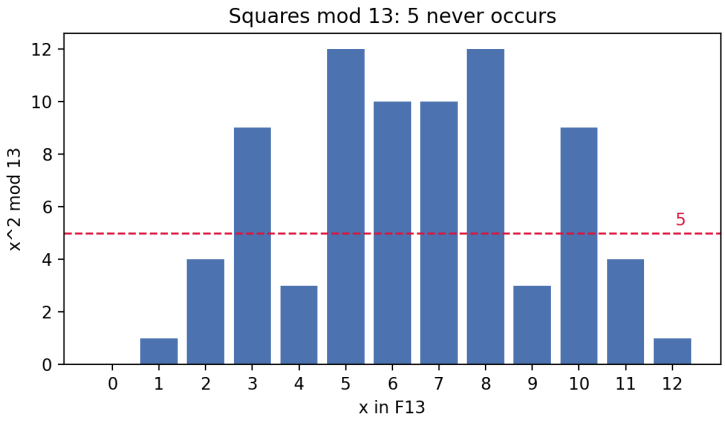


Figure 1: The squares mod 13. The value 5 never occurs, which is the condition needed for $\mathbb{F}_{13}[\theta]/(\theta^2 - 5)$ to be a field.

Table 1: The 28-cycle of powers of $\varphi = 7 + 6\theta$ in $\mathbb{F}_{169}$, read column by column.

k	φ^k	k	φ^k	k	φ^k	k	φ^k
0	1	7	8	14	12	21	5
1	$7 + 6\theta$	8	$4 + 9\theta$	15	$6 + 7\theta$	22	$9 + 4\theta$
2	$8 + 6\theta$	9	$12 + 9\theta$	16	$5 + 7\theta$	23	$1 + 4\theta$
3	$2 + 12\theta$	10	$3 + 5\theta$	17	$11 + 1\theta$	24	$10 + 8\theta$
4	$10 + 5\theta$	11	$2 + 1\theta$	18	$3 + 8\theta$	25	$11 + 12\theta$
5	$12 + 4\theta$	12	$5 + 6\theta$	19	$1 + 9\theta$	26	$8 + 7\theta$
6	$9 + 9\theta$	13	$7 + 7\theta$	20	$4 + 4\theta$	27	$6 + 6\theta$

3 Structure of the 28-cycle

Exactly four powers of φ have zero θ-component: 1, 8, 12, 5 at exponents 0, 7, 14, 21. These four elements form the unique order-4 subgroup of the order-28 cyclic group generated by φ. A cyclic group of order $28 = 4 \times 7$ splits into $28/4 = 7$ cosets of that subgroup: the subgroup itself, plus six further cosets of four elements each (Table 2).

4 A finite-field analogue of the 600-cell

The classical real 600-cell has 120 vertices built from three coordinate patterns: 8 permutations of $(\pm 1, 0, 0, 0)$, 16 vertices $(\pm\frac{1}{2})^4$, and 96 even permutations of $(\pm\frac{\varphi}{2}, \pm\frac{1}{2}, \pm\frac{\varphi-1}{2}, 0)$. We use the same recipe, with every constant evaluated in $\mathbb{F}_{13}[\theta]$ using modular inverses:

$$\tfrac{1}{2} = 7, \qquad \tfrac{\varphi}{2} = 10 + 3\theta, \qquad \tfrac{\varphi-1}{2} = 3 + 3\theta.$$

Generating all 120 vectors from this recipe and checking for coincidences gives exactly 120 distinct points, every one of squared norm $\langle v, v\rangle = 1$. Computing the pairwise inner product of one fixed vertex against all 120 (including itself) gives exactly nine distinct values in $\mathbb{F}_{13}[\theta]$; Table 3 shows that these nine values correspond exactly, one for one, to the nine real H_4 cosine values $\{\pm 1, \pm\varphi/2, \pm\frac{1}{2}, \pm(\varphi - 1)/2, 0\}$. Figure 2 shows the full 120×120 table of these inner products.

Table 2: The six further cosets of $\{1, 8, 12, 5\}$ inside $\langle\varphi\rangle$. Coset k is $\{\varphi^k, \varphi^{k+7}, \varphi^{k+14}, \varphi^{k+21}\}$.

Coset	Elements
1	$7+6\theta,\ 4+9\theta,\ 6+7\theta,\ 9+4\theta$
2	$8+6\theta,\ 12+9\theta,\ 5+7\theta,\ 1+4\theta$
3	$2+12\theta,\ 3+5\theta,\ 11+1\theta,\ 10+8\theta$
4	$10+5\theta,\ 2+1\theta,\ 3+8\theta,\ 11+12\theta$
5	$12+4\theta,\ 5+6\theta,\ 1+9\theta,\ 8+7\theta$
6	$9+9\theta,\ 7+7\theta,\ 4+4\theta,\ 6+6\theta$

Table 3: The nine-element inner-product spectrum, matched to the real H_4 cosine values.

$\mathbb{F}_{13}[\theta]$ value	Real H_4 value
0	0
1	1
$3+3\theta$	$(\varphi-1)/2$
$3+10\theta$	$-\varphi/2$
6	$-1/2$
7	$1/2$
$10+3\theta$	$\varphi/2$
$10+10\theta$	$-(\varphi-1)/2$
12	-1

5 The conjugate root ψ and the explicit mirror map

The second root of $x^2 - x - 1 = 0$, $\psi = 7 + 7\theta$, is the image of φ under the unique nontrivial automorphism of $\mathbb{F}_{169}$ over $\mathbb{F}_{13}$. This automorphism, the Frobenius map $x \mapsto x^{13}$, coincides exactly with coordinate conjugation $a + b\theta \mapsto a - b\theta$ on every element of the field, and in particular $\varphi^{13} = (7+6\theta)^{13} = 7+7\theta = \psi$. The two roots satisfy the trace and norm identities of $x^2 - x - 1 = 0$ exactly,

$$\varphi + \psi = 1, \qquad \varphi\psi = 12 = -1 \text{ in } \mathbb{F}_{13},$$

matching the elementary symmetric functions of the polynomial. Both roots have order 28, and

$$\frac{\psi}{2} = 10 + 10\theta = -\frac{\varphi - 1}{2}, \qquad \frac{\psi - 1}{2} = 3 + 10\theta = -\frac{\varphi}{2},$$

each an immediate consequence of $\psi = 1 - \varphi$, already present in Table 3.

Replacing φ by ψ in the 96-vertex recipe of Section 4 again yields a valid configuration of 120 norm-1 vectors with the identical nine-element spectrum, forced since Frobenius sends the φ-configuration's inner products to their own conjugates. The two 120-point sets are not identical: they share exactly the 24 rational vertices of type $(\pm 1, 0, 0, 0)$ and $(\pm\frac{1}{2})^4$, the 24-cell core of Section 8, while the remaining 96 vertices in each are genuinely different.

The key identity is $\psi = -1/\varphi$ exactly in $\mathbb{F}_{169}$, an immediate consequence of $\varphi\psi = -1$ together with $\varphi + \psi = 1$. Write the three nonzero magnitudes used in Section 4 as $v_1 = \varphi/2$, $v_2 = 1/2$, $v_3 = (\varphi - 1)/2$. Since $\psi/2 = -v_3$ and $(\psi - 1)/2 = -v_1$, the sets $\{\pm v_3\} = \{\pm\psi/2\}$ and $\{\pm v_1\} = \{\pm(\psi - 1)/2\}$ coincide exactly; only the sign labelling differs. So the ψ-recipe uses the same three unordered magnitudes as the φ-recipe, placed in swapped slots.

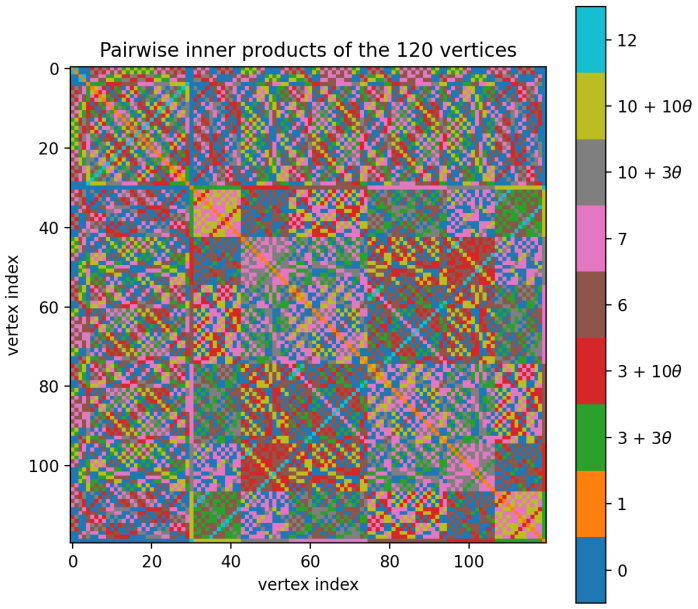


Figure 2: Exact pairwise inner products of the 120 finite-field vertices, independently recomputed. The nine colours are the nine values of Table 3.

Direct enumeration confirms what this algebra predicts. The 96 even and 96 odd permutations of $(v_1, v_2, v_3, 0)$, with all sign choices, are disjoint and together account for the full 192-element orbit of $(v_1, v_2, v_3, 0)$ under the hyperoctahedral group, and the 96 odd permutations coincide exactly, vertex for vertex, with the 96 vertices produced by the ψ-recipe. Consequently the linear map

$$T : (x_1, x_2, x_3, x_4) \mapsto (x_3, x_2, x_1, x_4),$$

the coordinate transposition swapping axes 1 and 3, an orthogonal involution of determinant -1, fixes the shared 24-point rational core setwise and carries the 96 φ-vertices exactly onto the 96 ψ-vertices. Checked directly against every one of the 120 vertices,

$$T(\varphi\text{-configuration}) = \psi\text{-configuration exactly, vertex for vertex.}$$

T is not one of the 14400 elements of the reflection group of Section 7: that group consists of the self-symmetries of a single configuration, while T is a congruence between the two distinct configurations, and direct check confirms it does not lie in that group.

5.1 Why conjugation is Frobenius, for any choice of θ

The check above was carried out for the specific choice $\theta^2 = 5$. The same conclusion holds for any nonresidue in place of 5, for a short reason. The six nonzero quadratic nonresidues modulo 13 are $\{2, 5, 6, 7, 8, 11\}$, the complement of $\{1, 3, 4, 9, 10, 12\}$ among $\{1, \ldots, 12\}$. For every one of these six values d, Euler's criterion gives $d^6 \equiv -1 \pmod{13}$, and this is all that is needed: if $\theta^2 = d$ for any such d, then

$$\theta^{13} = \theta^{12}\theta = (\theta^2)^6\theta = d^6\theta = -\theta,$$

so the Frobenius map $x \mapsto x^{13}$ sends $a + b\theta \mapsto a - b\theta$ regardless of which nonresidue defines θ. Consequently $\mathbb{F}_{13}[\theta]/(\theta^2 - d)$ is the same field $\mathbb{F}_{169}$, up to isomorphism, for every choice of d among these six values.

5.2 The p-adic origin of $\mathbb{F}_{169}$

This is not a coincidence of finite-field arithmetic alone. For the 13-adic field $\mathbb{Q}_{13}$, a quadratic extension generated by the square root of a unit that is a nonresidue modulo 13 is unramified, and $\mathbb{Q}_{13}$ has exactly one unramified quadratic extension. So

$$\mathbb{Q}_{13}(\sqrt{2}) = \mathbb{Q}_{13}(\sqrt{5}) = \mathbb{Q}_{13}(\sqrt{6}) = \mathbb{Q}_{13}(\sqrt{7}) = \mathbb{Q}_{13}(\sqrt{8}) = \mathbb{Q}_{13}(\sqrt{11})$$

are all the same unramified quadratic extension of $\mathbb{Q}_{13}$, and $\mathbb{F}_{169}$ is exactly its residue field, with θ the reduction of any one of these six square roots. Since $21 \equiv 8 \pmod{13}$ and $8 = 2 \cdot 2^2$ is 2 times a square, $\mathbb{Q}_{13}(\sqrt{21}) = \mathbb{Q}_{13}(\sqrt{8}) = \mathbb{Q}_{13}(\sqrt{2})$ as well, and globally $\mathbb{Q}(\sqrt{2}, \sqrt{21}) = \mathbb{Q}(\sqrt{2}, \sqrt{42})$ is the biquadratic field with Galois group $C_2 \times C_2$ whose three quadratic subfields are $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{21})$, and $\mathbb{Q}(\sqrt{42})$, since $2 \cdot 21 = 42$. Every finite-field computation in this note is the reduction modulo 13 of a computation that already takes place in this fixed unramified extension of $\mathbb{Q}_{13}$.

> **Lemma.** $\mathbb{Q}_{13}(\sqrt{21}) = \mathbb{Q}_{13}(\sqrt{2})$, the unique unramified quadratic extension of $\mathbb{Q}_{13}$. Consequently $\mathbb{Q}(\sqrt{42}, \sqrt{2}) = \mathbb{Q}(\sqrt{2}, \sqrt{21})$, a field of degree four over $\mathbb{Q}$ with Galois group $C_2 \times C_2$, and the Frobenius substitution acts on the residue field by $\theta \mapsto -\theta$, exchanging the two conjugate eigenvalues of the generator.
>
> *Proof.* The squares modulo 13 are $\{0, 1, 3, 4, 9, 10, 12\}$; neither 2 nor $21 \equiv 8$ lies among them, so both generate quadratic extensions. Since $8 = 2 \cdot 4$ and 4 is a square, the two extensions coincide, giving $\mathbb{Q}_{13}(\sqrt{21}) = \mathbb{Q}_{13}(\sqrt{2})$; a quadratic extension of $\mathbb{Q}_{13}$ generated by a unit that is a nonresidue is unramified, and there is exactly one such extension. The degree-four identity holds because $42 = 2 \cdot 21$, so each of the two fields contains the same three quadratic subfields, and the degree is four because $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{21})$, $\mathbb{Q}(\sqrt{42})$ are distinct. On the residue field the Frobenius is $x \mapsto x^{13}$, and $\theta^{13} = \theta^{12}\theta = (\theta^2)^6\theta = 2^6\theta = 64\theta \equiv 12\theta = -\theta$, so it sends $10 + 7\theta$ to $10 - 7\theta$, exchanging the conjugate pair.

6 The golden 28-cycle inside the primitive root γ

Write $\gamma = 7 + 8\theta$ for the element rejected in Section 2, the primitive root of the full group $\mathbb{F}_{169}^{\times}$ of order 168. Since $\langle\gamma\rangle$ is cyclic of order 168, it has a unique subgroup of order 28, namely $\langle\gamma^6\rangle$ (as $168/28 = 6$); and since φ and ψ both have order 28, each is automatically a generator of this same subgroup:

$$\langle\varphi\rangle = \langle\psi\rangle = \langle\gamma^6\rangle \subset \langle\gamma\rangle = \mathbb{F}_{169}^{\times}.$$

Figure 3 shows the nested subgroup structure. An independent SymPy computation, using polynomial arithmetic over $GF(13)$ reduced modulo $\theta^2 - 5$, finds the exact discrete logarithms

$$\varphi = \gamma^{30} = (\gamma^6)^5, \qquad \psi = \gamma^{54} = (\gamma^6)^9, \qquad \gcd(5, 28) = \gcd(9, 28) = 1,$$

confirming that both exponents are coprime to 28 and so both φ and ψ generate the full order-28 subgroup. The same check gives

$$\varphi^{-1} = \gamma^{138} = (\gamma^6)^{23}, \qquad -1 = \varphi^{14} = \gamma^{84} = (\gamma^6)^{14},$$

the second identity holding because φ^{14} is the unique element of order dividing 2 in $\langle\varphi\rangle$, and the only such element in any field is -1.

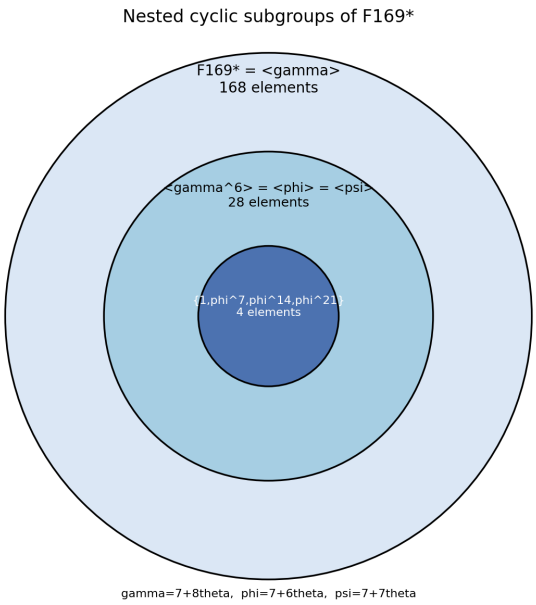


Figure 3: The nested cyclic subgroups of $\mathbb{F}_{169}^{\times}$: the full 168-element group generated by γ, its unique 28-element subgroup generated equally by φ or ψ, and the order-4 subgroup inside that.

The golden coordinates of Section 4 can be read directly off the γ-cycle. Before the common factor $1/2$, the three nonzero magnitudes and the sign -1 used to build them are all powers of γ:

$$1 = \gamma^0, \qquad \varphi = \gamma^{30}, \qquad \varphi^{-1} = \gamma^{138}, \qquad -1 = \gamma^{84},$$

so that $\frac{1}{2}(\pm\varphi, \pm 1, \pm\varphi^{-1}, 0) = \frac{1}{2}(\pm\gamma^{30}, \pm\gamma^{0}, \pm\gamma^{138}, 0)$, with every sign realized by multiplying by $\gamma^{84} = -1$. The factor $1/2 = 7$ itself lies in $\mathbb{F}_{13}^{\times}$, the order-12 subgroup of elements with zero θ-component, and $\gcd(28, 12) = 4 \neq 28$, so $\langle\gamma^6\rangle$ and $\mathbb{F}_{13}^{\times}$ meet only in the order-4 subgroup $\{1, \gamma^{6\cdot 7}, \gamma^{6\cdot 14}, \gamma^{6\cdot 21}\} = \{\varphi^0, \varphi^7, \varphi^{14}, \varphi^{21}\}$ of Table 2; the overall scaling by $1/2$ is a separate external factor.

7 Genuine reflections and the full H_4 group

Because every vertex constructed above has norm 1, a reflection through any vertex u,

$$S_u(v) = v - 2\langle v, u\rangle u,$$

needs no division and no square root beyond θ itself. For an arbitrary vertex u, direct checking (all 120 vertices, not a sample) confirms that S_u is an involution, preserves the norm, and maps the 120-point set onto itself.

An exhaustive search over the 120 vertices locates a chain u_1, u_2, u_3, u_4 whose pairwise inner products match the H_4 Dynkin diagram exactly (Figure 4):

$$u_1 = (0, 0, 0, 1),\ u_2 = (0, 3{+}10\theta, 3{+}3\theta, 6),\ u_3 = (3{+}3\theta, 10{+}3\theta, 7, 0),\ u_4 = (6, 10{+}10\theta, 3{+}10\theta, 0),$$

with exact Gram matrix

$$\big(\langle u_i, u_j\rangle\big)_{i,j=1}^{4} = \begin{pmatrix} 1 & 6 & 0 & 0 \\ 6 & 1 & 6 & 0 \\ 0 & 6 & 1 & 3{+}10\theta \\ 0 & 0 & 3{+}10\theta & 1 \end{pmatrix}.$$

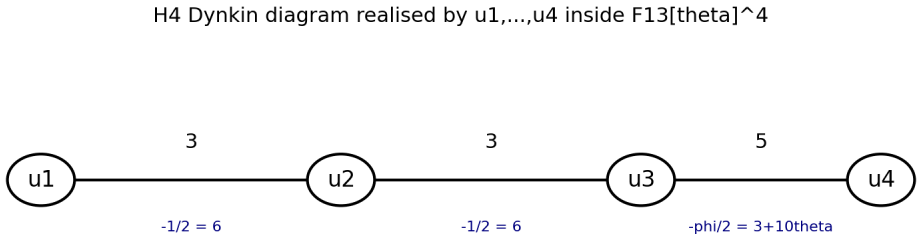


Figure 4: The H_4 Dynkin diagram realised by the simple-root chain $u_1, \ldots, u_4$.

The four reflection matrices $S_1, \ldots, S_4$ each satisfy $S_i^2 = I$ exactly over $\mathbb{F}_{13}[\theta]$. Enumerating the group they generate by breadth-first search over 4×4 matrices, with no truncation, gives a group of order 14400, identical to the order of the real H_4 Coxeter group (Figure 5). This is the central result of this note: H_4 has a faithful realization entirely inside $GL(4, \mathbb{F}_{13}[\theta])$, built without any real or complex number.

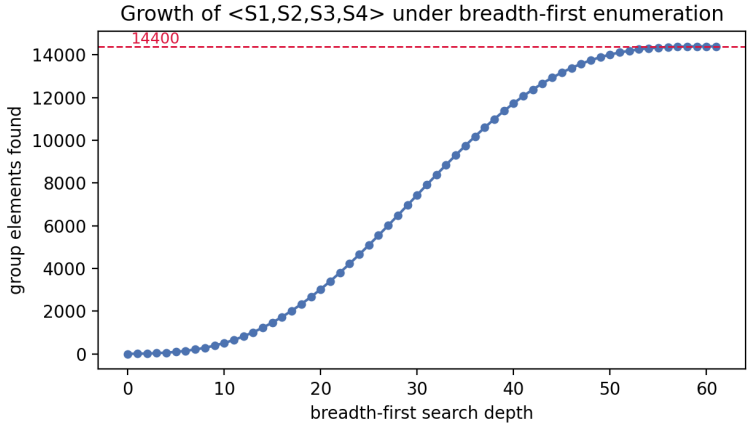


Figure 5: Breadth-first enumeration of $\langle S_1, S_2, S_3, S_4\rangle$: the number of distinct matrices found saturates at exactly 14400.

8 The 24-cell

The 24 vertices of type $(\pm 1, 0, 0, 0)$ and $(\pm\frac{1}{2})^4$ have pairwise inner-product spectrum exactly the five-element set

$$\{1,\ \tfrac{1}{2},\ 0,\ -\tfrac{1}{2},\ -1\} \longleftrightarrow \{1,\ 7,\ 0,\ 6,\ 12\},$$

the sub-list of Table 3 not involving φ, matching the real 24-cell's edge-equals-circumradius angle set exactly.

9 T_3: the integer companion matrix and its Lagrange-Sylvester projectors

This section attaches the integer Tribonacci companion matrix T_3 to the field-theoretic picture above. Everything is computed over $\mathbb{Z}$, then $\mathbb{F}_{13}$, then $\mathbb{Z}_{13}$, with no real or complex number and no floating-point arithmetic at any stage (Figure 6).

Let

$$T_{\mathbb{Z}} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in SL(3, \mathbb{Z}),$$

the companion matrix of $x^3 - x^2 - x - 1$, with $\det T_{\mathbb{Z}} = 1$ over $\mathbb{Z}$. Its reduction $T \in SL(3, \mathbb{F}_{13})$ has order 168 and characteristic polynomial factoring as

$$x^3 - x^2 - x - 1 \equiv (x + 6)(x^2 + 6x + 2) \pmod{13}.$$

The rational discriminant of $x^3 - x^2 - x - 1$ is $-44 \neq 0$, the factorization above is squarefree, so T is semisimple.

9.1 Lagrange-Sylvester idempotents by Bezout

Write $f_1(x) = x - 7$, $f_2(x) = x^2 + 6x + 2$ (using $-6 \equiv 7$), so $x^3 - x^2 - x - 1 \equiv f_1(x) f_2(x)$. Since $f_2(7) = 93 \equiv 2 \pmod{13}$ and $2^{-1} \equiv 7 \pmod{13}$, the constant polynomial $u(x) = 7$ satisfies $f_2(x)u(x) \equiv 1 \pmod{f_1(x)}$, giving the Bezout idempotents

$$e_1(x) = 7 f_2(x) = 7x^2 + 3x + 1, \qquad e_2(x) = 1 - e_1(x) = 6x^2 + 10x \pmod{13}.$$

Evaluating at T gives the exact Lagrange-Sylvester projectors, over $\mathbb{F}_{13}$,

$$P_1 = 7T^2 + 3T + I = \begin{pmatrix} 1 & 7 & 10 \\ 3 & 8 & 4 \\ 7 & 10 & 5 \end{pmatrix}, \qquad P_{23} = 6T^2 + 10T = \begin{pmatrix} 0 & 6 & 3 \\ 10 & 6 & 9 \\ 6 & 3 & 9 \end{pmatrix}.$$

Direct computation confirms, without exception: $P_1^2 = P_1$, $P_{23}^2 = P_{23}$, $P_1 P_{23} = P_{23} P_1 = 0$, $P_1 + P_{23} = I$, $\operatorname{rank}(P_1) = 1$, $\operatorname{rank}(P_{23}) = 2$, $TP_1 = 7P_1$, $(T^2 + 6T + 2I)P_{23} = 0$, and $7P_1 + TP_{23} = T$ exactly.

9.2 Rank-one structure of P_1

P_1 is the outer-product projector built from the right eigenvector $v_0 = (1, 3, 7)$, with $Tv_0 = 7v_0$, and the left eigenvector $w_0 = (4, 2, 1)$, with $w_0 T = 7w_0$:

$$P_1 = \frac{v_0 w_0^{\top}}{w_0 \cdot v_0}, \qquad w_0 \cdot v_0 = 4,$$

and this outer-product form reproduces the Bezout-derived P_1 exactly, entry for entry.

9.3 The quadratic block identifies the same field $\mathbb{F}_{169}$

Completing the square on f_2: $(T + 3I)^2 P_{23} = 7P_{23}$ exactly. So on the rank-2 image of P_{23} the matrix T behaves as a root of $\theta^2 - 7$. Since $7 \in \{2, 5, 6, 7, 8, 11\}$, the six nonresidues modulo 13 identified above, this is, up to isomorphism, the very same field $\mathbb{F}_{169}$ built from $\theta^2 - 5$ and used throughout the H_4/600-cell construction: the centralizer extension of T_3 and the golden-ratio field are one and the same object.

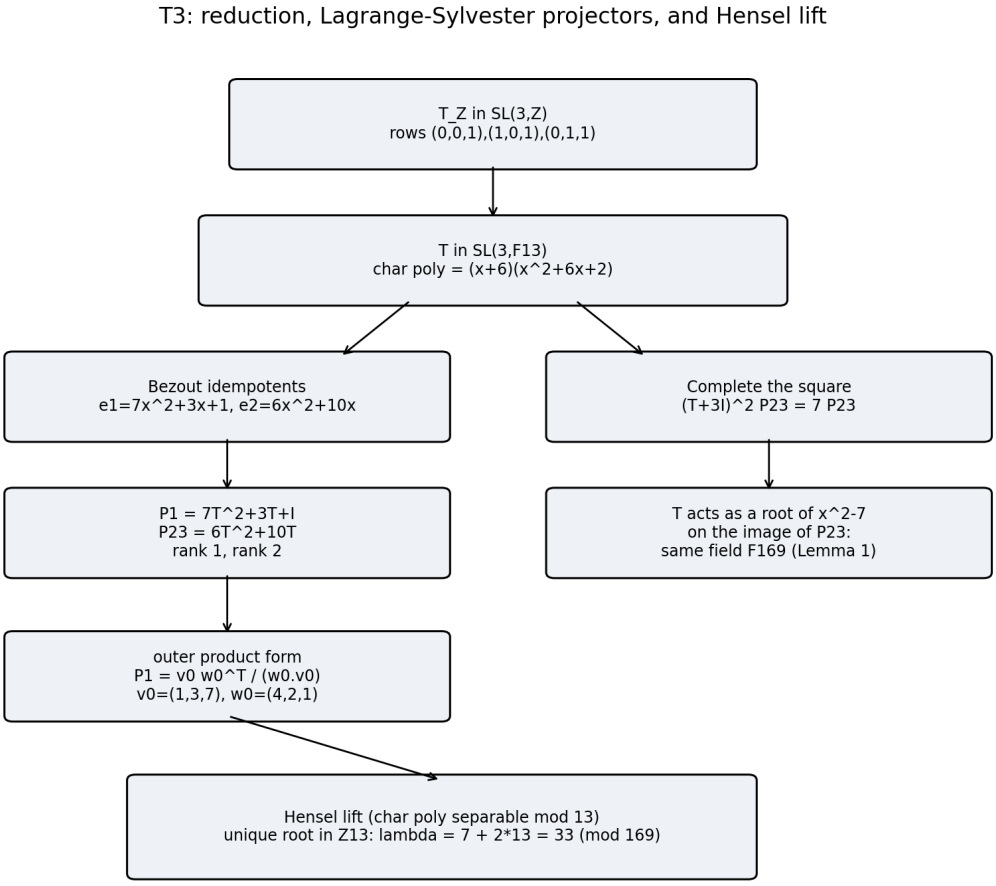


Figure 6: T_3 reduced mod 13, its Bezout/CRT Lagrange-Sylvester projectors, the rank-one outer-product form of P_1, and the Hensel lift of the eigenvalue to $\mathbb{Z}_{13}$.

9.4 The 13-adic lift, via Hensel's lemma

Because $x^3 - x^2 - x - 1$ is separable mod 13 (its discriminant $-44 \equiv 8 \pmod{13}$ is nonzero, and $f'(7) = 132 \equiv 2 \pmod{13}$ is nonzero at the root), Hensel's lemma applies directly to the integer polynomial: the factorization $(x-7)(x^2+6x+2)$ lifts uniquely to a factorization of $x^3 - x^2 - x - 1$ over $\mathbb{Z}_{13}$, with the linear factor's root a genuine 13-adic integer $\lambda \in \mathbb{Z}_{13}$, $\lambda \equiv 7 \pmod{13}$. One Newton-Hensel step lifts this to modulus $13^2 = 169$: solving $f(7) + 13k f'(7) \equiv 0 \pmod{169}$ with $f(7) = 286$, $f'(7) = 132$ gives $k = 2$, hence

$$\lambda \equiv 7 + 2 \cdot 13 = 33 \pmod{169}, \qquad f(33) = 34814 = 169 \times 206,$$

confirmed by exact integer division. T_3's spectral decomposition $P_1 \oplus P_{23}$ is not merely a mod-13 artifact: it is the reduction of a genuine decomposition of $T_{\mathbb{Z}}$ over the 13-adic integers $\mathbb{Z}_{13} \times \mathcal{O}_K$, where $\mathcal{O}_K$ is the ring of integers of the unramified quadratic extension $K = \mathbb{Q}_{13}(\sqrt{2}) = \mathbb{Q}_{13}(\sqrt{21})$ established above.

10 Order taxonomy of $\mathbb{F}_{169}^{\times}$

The group $\mathbb{F}_{169}^{\times}$ has 168 elements. We classify every one of them by multiplicative order and find sixteen classes, one for each divisor of 168 (Figure 7), with the count in each class equal to Euler's totient of that divisor.

Table 4: The sixteen order classes of $\mathbb{F}_{169}^{\times}$, with one representative each. The six smallest orders are exactly the rational elements (in $\mathbb{F}_{13}$); every order above 12, apart from the value 12 itself, requires θ.

order	count	representative
1	1	1
2	1	12 ($= -1$)
3	2	3
4	2	5 ($= i$, since $5^2 = -1$)
6	2	4
7	6	$4 + 4\theta$
8	4	θ
12	4	2 (and the rational T_3 eigenvalue 7)
14	6	$3 + 5\theta$
21	12	$2 + 5\theta$
24	8	2θ (a square root of 2)
28	12	φ, ψ
42	12	$1 + \theta$
56	24	$1 + 3\theta$
84	24	$2 + 2\theta$
168	48	γ (and the complex T_3 eigenvalues)

Two facts organize this table. First, since $13 \equiv 1 \pmod 4$, Euler's criterion makes -1 a quadratic residue mod 13, and indeed $5^2 = 25 \equiv -1 \pmod{13}$: the element 5 already plays the role of i inside $\mathbb{F}_{13}$ itself, no extension needed. Second, $\theta^2 = 5$, so $\theta^4 = 5^2 = -1$ and $\theta^8 = 1$: θ is a primitive 8th root of unity, literally a square root of i. The extension $\mathbb{F}_{13}[\theta]$ is needed only because -1, while already a square in $\mathbb{F}_{13}$, has no square root of its own square root there; $168 = 8 \times 3 \times 7$, and 8 is exactly the 2-part of the group order that forces this extension.

The order-12 elements are precisely the elements of $\mathbb{F}_{13}^{\times}$: since $\mathbb{F}_{169}^{\times}$ is cyclic of order 168 and $\mathbb{F}_{13}^{\times}$ has order 12, and 12 divides 168, $\mathbb{F}_{13}^{\times}$ is the unique subgroup of order 12, so every element of order dividing 12 is rational and every rational element has order dividing 12. All orders above this list (7, 8, 14, 21, 24, 28, 42, 56, 84, 168) accordingly require θ.

11 A single generator for T_3 and φ

The Tribonacci companion matrix T_3 has characteristic polynomial $x^3 - x^2 - x - 1 \equiv (x - 7)(x^2 + 6x + 2) \pmod{13}$, with roots

$$\lambda_1 = 7, \qquad \lambda_2 = 10 + 11\theta, \qquad \lambda_3 = 10 + 2\theta,$$

all three inside $\mathbb{F}_{169}$: no cubic extension is needed, since the quadratic factor's discriminant is $28 \equiv 2 \pmod{13}$, a nonresidue, and its roots already lie in $\mathbb{F}_{169}$. These satisfy the expected symmetric

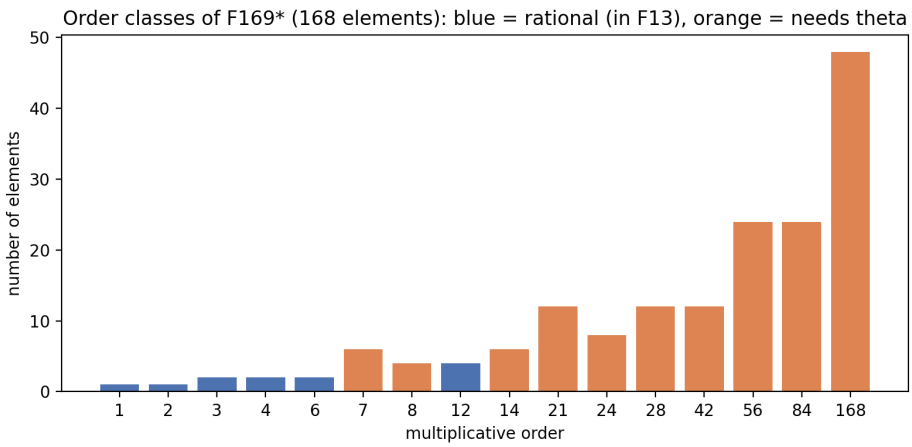


Figure 7: The sixteen order classes of $\mathbb{F}_{169}^{\times}$. Blue bars are rational elements (lying in $\mathbb{F}_{13}$); orange bars require θ.

functions $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and $\lambda_1\lambda_2\lambda_3 = 1$. A second, independent Lagrange-Sylvester decomposition, computed directly from these three eigenvalues rather than from the Bezout idempotents of Section 9, reproduces P_1 and $P_{23} = P_2 + P_3$ exactly, entry for entry.

Using $\gamma = 7 + 8\theta$ as reference generator, an independent SymPy computation gives the exact discrete logarithms of every named element in this note (Figure 8):

$$\lambda_1 = \gamma^{154}, \quad \lambda_2 = \gamma^{121}, \quad \lambda_3 = \gamma^{61}, \quad \varphi = \gamma^{30}, \quad \psi = \gamma^{54}, \quad \theta = \gamma^{147}, \quad 5 = \gamma^{126}.$$

The Frobenius relation is visible directly on the exponents: $13 \times 121 = 61 \pmod{168}$ and $13 \times 30 = 54 \pmod{168}$, so multiplication of the exponent by 13 exchanges each conjugate pair (λ_2, λ_3) and (φ, ψ), consistent with $13^2 \equiv 1 \pmod{168}$, an involution. So T_3 and φ are not merely elements of the same field: their entire algebraic data, eigenvalues, golden ratio, and the extension generator itself, are explicit powers of one shared generator.

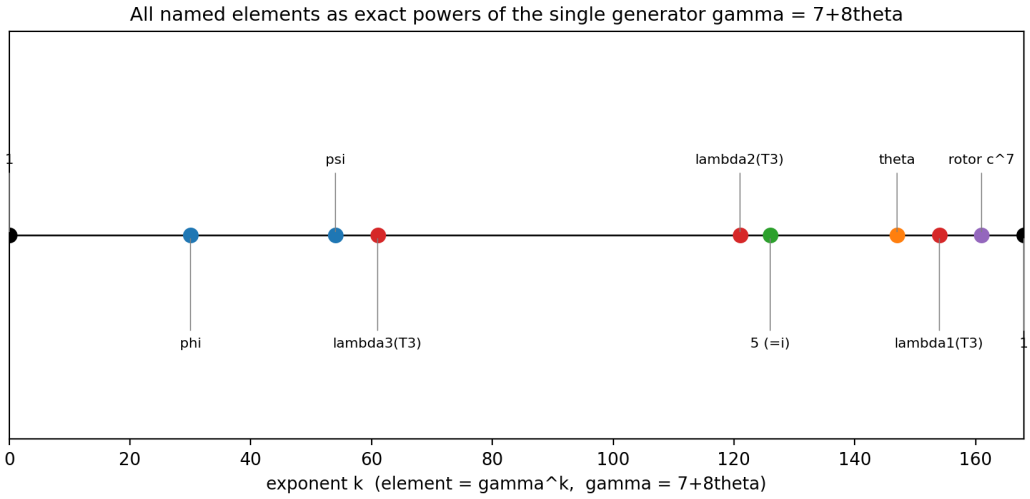


Figure 8: Every named element of this note as an explicit power of $\gamma = 7 + 8\theta$.

12 E_8, the golden 600-cells φH_4 and ψH_4, and the 24-cell

This section names the two 120-point configurations of Sections 4 and 5 explicitly as φH_4 and ψH_4, identifies each as a genuine quaternion group, and connects the resulting picture to the exceptional root system E_8. Two kinds of claim appear below. Verified facts were checked here, independently, by exact computation over $\mathbb{Q}(\varphi)$ (rational arithmetic on pairs (a, b) with $\varphi^2 = \varphi + 1$, no floating point). Cited facts are classical theorems recalled for context, not re-derived in full here.

Write φH_4 for the 120-point configuration built from φ and ψH_4 for the Galois-conjugate configuration built from ψ. Both carry a faithful action of the Coxeter group H_4 of order 14400.

Verified: φH_4 and ψH_4 are the binary icosahedral group $2I$. Reading each 4-tuple (w, x, y, z) as a quaternion $w + xi + yj + zk$ with coefficients in the exact ring $\mathbb{Z}[\varphi]$, the following were checked directly, with no sampling: every one of the 120 vertices of φH_4 has quaternion norm exactly 1; the set is closed under quaternion multiplication, all $120 \times 120 = 14400$ products land back inside the same 120-element set, with zero exceptions. The same two facts hold for ψH_4. A finite set of unit quaternions closed under multiplication is a finite subgroup of the unit quaternions, so both φH_4 and ψH_4 are genuine groups of order 120. The only finite subgroup of the unit quaternions of order 120 is the binary icosahedral group, so

$$\varphi H_4 \cong \psi H_4 \cong 2I \quad \text{(binary icosahedral group, order 120)}.$$

Verified: the shared 24-cell is the binary tetrahedral group $2T$. The 24 rational vertices shared by φH_4 and ψH_4 were tested the same way: all $24 \times 24 = 576$ pairwise quaternion products land back inside the same 24-element set, with zero exceptions. This 24-element quaternion group is, up to the standard identification, the binary tetrahedral group $2T$ of order 24, sitting inside both φH_4 and ψH_4 as a genuine subgroup (Figure 9).

Cited: the E_8 connection. It is a classical fact (the icosian construction of E_8: Conway and Sloane, *Sphere Packings, Lattices and Groups*, Ch. 8; the golden H_4 folding of E_8: Elser and Sloane 1987; Moody and Patera 1993) that the 240-root E_8 root system in $\mathbb{R}^8$ decomposes, under a linear projection built from the golden ratio, into exactly two orbits of 120 roots, each a 600-cell carrying the binary icosahedral group $2I$, related to one another by the same Galois conjugation $\varphi \leftrightarrow \psi = 1 - \varphi$ used throughout this note. In this classical picture the two orbits are, up to overall scale, precisely φH_4 and ψH_4 as verified above, and the shared 24-cell corresponds to the D_4 root system sitting inside E_8. This connection is recalled for context and is not independently re-derived in the full 8-dimensional setting here; what is verified above is that φH_4 and ψH_4 are exact realizations of $2I$ with a shared $2T$ core, the group-theoretic content the E_8 folding requires of its two halves.

13 Reproducibility

All arithmetic in this note is exact modular arithmetic on integer pairs (a, b) mod 13, implemented directly in Python with no external number library, cross-checked independently in SymPy for the discrete-log results of Sections 7 and 9. The only searches performed, the simple-root chain of Section 7 and the 14400-element group enumeration, are exhaustive breadth-first searches with an explicit termination check, not bounded approximations. The mirror map T of Section 5 was checked against all 120 vertices directly, and its non-membership in the 14400-element group was checked against all 14400 enumerated matrices directly. The 120-vertex configuration, its nine-value inner-product spectrum, the H_4 simple-root chain and Gram matrix, and the order-14400 group enumeration were independently recomputed from scratch, and all results agreed exactly with

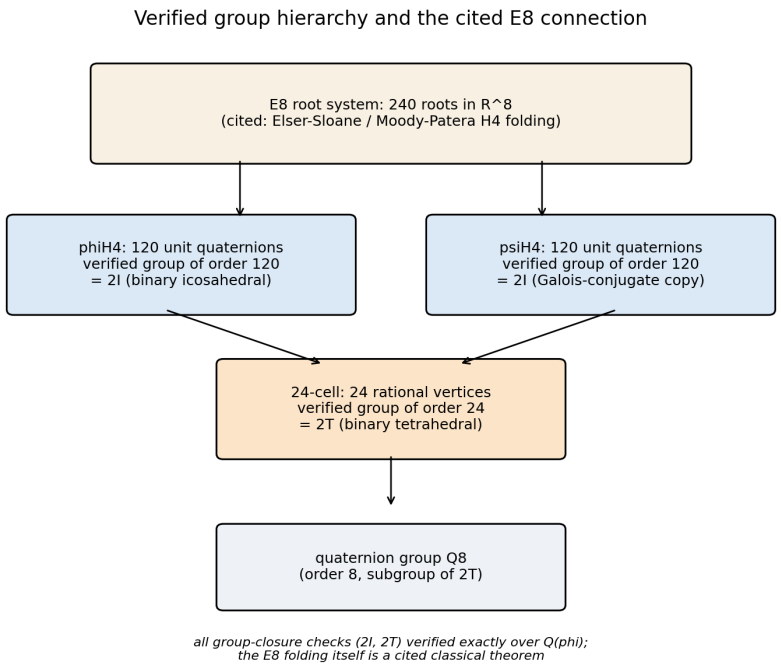


Figure 9: The verified group hierarchy: $\varphi H_4, \psi H_4 \cong 2I$ (order 120), sharing a 24-cell core $\cong 2T$ (order 24), and the classical, cited H_4-folding placing $\varphi H_4 \cup \psi H_4$ inside the E_8 root system.

those stated above. The Lagrange-Sylvester projectors of Section 9, the rank and orthogonality checks, the outer-product form, the field identification, and the Hensel lift to modulus 13^2 were likewise computed independently in Python and SymPy using exact integer arithmetic throughout, with no floating-point number appearing at any stage. The order taxonomy of Section 10 was obtained by an exhaustive scan of all 168 nonzero field elements, and every discrete logarithm reported was verified by direct exponentiation, not inferred from order alone. The group-closure checks (Section 10, φH_4, ψH_4, and the shared 24-point core) were performed over $\mathbb{Q}(\varphi)$ using exact rational arithmetic (Python's `fractions.Fraction`), with every one of the 120×120, 120×120, and 24×24 pairwise quaternion products checked directly and no floating-point number appearing at any stage; the E_8-folding claim itself is cited, not independently re-derived in eight dimensions here.